Journal of Electrical Engineering & Technology (2026) 21:217–231
https://doi.org/10.1007/s42835-025-02443-8

ORIGINAL ARTICLE

The Evaluation of Benefits from Green Electricity Trading in New Energy Power Stations

Bin Wang[1] · Shuailiang Yao[1] · Haixu Liu[1] · Dongxia Cheng[1] · Hongyong Liu[1] · Jian Kou[1] · Chengdong Yang[1] · Changjiang Su[1]

Received: 8 January 2025 / Revised: 7 August 2025 / Accepted: 8 September 2025 / Published online: 15 September 2025
© The Author(s) under exclusive licence to The Korean Institute of Electrical Engineers 2025

Abstract
With the growing global emphasis on climate change mitigation and sustainable development, renewable energy sources such as wind, solar, and hydro power have become essential for transforming electricity systems. To facilitate the effective integration of renewables into electricity markets, researchers have increasingly focused on developing optimized and transparent market mechanisms. However, existing evaluation frameworks for green power projects remain limited, often emphasizing economic outcomes while neglecting critical social and environmental dimensions, as well as system-level constraints. To address these limitations, this study develops a comprehensive benefit evaluation framework for green electricity trading in new energy stations, encompassing economic, environmental, and social indicators. The model integrates fuzzy comprehensive evaluation and an improved AHP-entropy weighting method, with added robustness through Monte Carlo simulation and sensitivity analysis of expert weights. To account for real-world constraints, the framework considers potential transmission congestion, locational grid access limitations, and external climate risks—such as prolonged low-wind periods or dust storms—via scenario-based simulations.The results from multiple case studies demonstrate that the proposed model performs more reliably and consistently than baseline approaches across most evaluation criteria. The inclusion of uncertainty bounds and statistical validation (e.g., RMSE, significance tests) further supports the credibility of the outcomes. This study concludes that green power projects offer strong multidimensional benefits and proposes practical policy recommendations to support the rational development of green electricity markets. The framework contributes meaningfully to advancing the large-scale, resilient deployment of renewable energy within complex power systems.

Keywords Green electricity trading · New energy stations · Assessment methods · Comprehensive benefits · Evaluation indicators

✉ Shuailiang Yao
1792420229@qq.com

Bin Wang
wang.bin.1@jibei.sgcc.com.cn

Haixu Liu
280469256@qq.com

Dongxia Cheng
cheng.dongxia@jibei.sgcc.com.cn

Hongyong Liu
liu_hongyong123@126.com

Jian Kou
392657204@qq.com

Chengdong Yang
1246252380@qq.com

Changjiang Su
308734902@qq.com

[1] State Grid Jibei Zhang Jiakou Wind, PV Storage and Transmission Renewable Energy CO., LTD, Zhangjiakou, China

1 Introduction

Global efforts to combat climate change and promote sustainable development have intensified the transition toward green electricity systems. Historically, energy structures evolved from wood and coal to oil and gas, fueling industrialization but also exacerbating environmental degradation and climate risks. In the twenty-first century, renewable technologies such as wind and solar energy have emerged as key alternatives [1]. These developments, coupled with digital and smart technologies, are reshaping global energy systems from centralized, fossil-based grids to decentralized, low-carbon networks.

Achieving carbon peaking and neutrality goals requires robust market mechanisms, modernized grid infrastructure, and effective policy incentives. Green electricity trading has become a vital strategy for governments and corporations to meet sustainability targets. China's implementation of green electricity trading mechanisms in several pilot regions—surpassing 50 billion kWh in transactions—demonstrates rapid market growth [2]. However, current project evaluations focus largely on economic metrics [3, 4], often neglecting environmental and social impacts.

A more comprehensive benefit evaluation framework is urgently needed. While both domestic and international scholars have developed innovative tools—such as fuzzy evaluation, AHP, neural networks, and DEA [5–22]—existing models still fall short in addressing real-world constraints. Notably, factors such as transmission congestion, grid access limitations, and extreme weather scenarios are rarely integrated, and uncertainty quantification is often missing.

This study proposes a multidimensional evaluation model for green electricity trading projects at new energy stations. The model combines fuzzy comprehensive evaluation with improved AHP-entropy weighting, integrates scenario-based analysis of grid and climate constraints, and applies Monte Carlo simulations and statistical validation for robustness. Using real-world case studies, the model aims to improve decision-making accuracy, enhance transparency, and support sustainable renewable energy development [23–25].

2 Benefit Indicator Model

This section introduces the model for economic indicators as well as the qualitative analysis of environmental and social indicators.

2.1 Economic Indicators

According to the Hebei Development and Reform Commission's "Notice on the Green Power Market Transaction Work for the North Hebei Power Grid in 2022" (Document No. 41 [2022] of Hebei Development and Reform Commission), green power trading is categorized into two types: guaranteed transactions and market transactions.

Specific requirements for the Beijing-Tianjin-Hebei green trading market involve allocating remaining channel capacity to the leftover units after meeting the electricity generation needs of photovoltaic poverty alleviation stations and guaranteed renewable energy stations. To develop an optimized strategy for units participating in the green power market, a trading model aimed at maximizing the revenue of new energy stations in green power trading should be established. The power generation input to the market is based on the actual data from the Hebei North New Energy Power Station.

$$\max \sum_{t=1}^{M} \left(\Delta t \cdot \pi^{l} + \Delta t' \cdot \pi^{h} \right) \tag{1}$$

Where $f(t)$ represents the power generation revenue for station i on day t; Δt denotes the third-priority market trading volume for station i on day t, and $\Delta t'$ denotes the fourth-priority market trading volume for station i on day t. The daily trading volume is the sum of the three segment volumes each day;π^{l} represents the trading price in the third-priority market, and π^{h} represents the trading price in the fourth-priority market; M is the number of days in the month.

Based on the differences in market conditions and policies across various regions, the rationality of electricity price setting is as follows:

$$V_P = \left[\left(P'_E - P_E \right) / P_E \right] / \left[\left(\frac{Q'_s}{Q'_d} - \frac{Q_s}{Q_d} \right) / \frac{Q_s}{Q_d} \right] \tag{2}$$

Where Formula (2) is Rationality of Electricity Pricing Method; P_E is market electricity price during the base period;P'_E is market electricity price during the reporting period;Q_s is total electricity supply during the base period; Q_d is total electricity demand during the base period; Q'_s is total electricity supply during the reporting period; Q'_d is total electricity demand during the reporting period. Formula $V_{an} = \frac{T_{an}}{T_T}$ is Price Stability of Electricity; T_{an} is duration of the period with abnormal electricity prices, denoted as T, divided by the total duration of the operating period. Formula $MEI = \frac{MC}{P}$ is Price Effectiveness P Transaction price.

Considering that large-scale electricity trading may lead to transmission congestion, the following optimization model is constructed based on nodal marginal pricing:

$$\min_{P_{\mathrm{G}i},\, P_j} \left[\sum_{i=1}^{M_{\mathrm{G}}} f_i\left(P_{\mathrm{G}i}\right) - \sum_{j=1}^{M_{\mathrm{L}}} f_j\left(P_{\mathrm{L}j}\right) \right] \tag{3}$$

$$\begin{cases} \sum_{i=1}^{M_{\mathrm{G}}} P_{\mathrm{G}i} - \sum_{j=1}^{M_{\mathrm{L}}} P_{\mathrm{L}j} = 0 \\ P_{\mathrm{G}i}^{\min} P_{\mathrm{G}i} P_{\mathrm{G}i}^{\max}, \quad i = 1, 2, \cdots, M_{\mathrm{G}} \\ P_{\mathrm{L}j}^{\min} P_{\mathrm{L}j} P_{\mathrm{L}j}^{\max}, \quad j = 1, 2, \cdots, M_{\mathrm{L}} \\ P_k^{\min} P_k P_k^{\max}, \quad k = 1, 2, \cdots, M_{\mathrm{K}} \end{cases} \tag{4}$$

$$f_i\left(P_{\mathrm{G}i}\right) = \frac{1}{2} U_{\mathrm{G}i} P_{\mathrm{G}i}^2 + V_{\mathrm{G}i} P_{\mathrm{G}i}, i = 1, 2, \cdots, M_{\mathrm{G}} \tag{5}$$

$$h_j\left(P_{\mathrm{L}j}\right) = -\frac{1}{2} U_{\mathrm{L}j} P_{\mathrm{L}j}^2 + V_{\mathrm{L}j} P_{\mathrm{L}j}, j = 1, 2, \cdots, M_{\mathrm{L}} \tag{6}$$

$$\zeta_{\mathrm{G}i} = \lambda - \sum_{k=1}^{N_{\mathrm{L}}} \left(\alpha_k + \beta_k\right) \frac{\partial P_k}{\partial P_{\mathrm{G}i}} \tag{7}$$

$$\zeta_{\mathrm{L}j} = \lambda + \sum_{k=1}^{N_{\mathrm{L}}} \left(\alpha_k + \beta_k\right) \frac{\partial P_k}{\partial P_{\mathrm{L}j}} \tag{8}$$

Where Eqs. (3) and (4) represent the objective function and constraint conditions, respectively. M_G, M_L, and M_K denote the number of generator nodes, load nodes, and transmission branches, respectively. P_{Gi}, P_{Lj}, and P_k represent the active power at generator node i, load node j, and branch k, respectively P_{Gi}^{max} and P_{Gi}^{min} are the upper and lower limits of active power at generator node i; P_{Lj}^{max} and P_{Lj}^{min} are the upper and lower limits of active power at load node j; P_k^{max} and P_k^{min} are the upper and lower limits of active power for branch k. $f_i\,()$ and $h_i\,(\cdot)$ denote the generation cost function for generator node i and the utility function for user j, respectively, as defined in Eqs. (5) and (6). U_{Gi} and V_{Gi} represent the bidding curve parameters of generator i in the electricity market, while U_{Lj} and V_{Lj} represent the bidding curve parameters of user j in the electricity market. According to the theory of real-time electricity pricing, the marginal nodal prices at generator node i and load node j, denoted by ζ_{Gi} and ζ_{Lj}, are calculated using Eqs. (7) and (8), respectively.Here, α_k and β_k are the Lagrange multipliers associated with the equality constraints of the optimization model and the power flow constraints of the branches.The partial derivative terms represent the sensitivity elements in the matrix that describes the impact of nodal power injections on transmission line flows.

Method for Determining the Transactional Electricity Volume in New Energy Stations [26]

(1) All Units Participate in the Tertiary Market

On the first d days of the month, all units in the market participate in the tertiary market, with equal priority for all members. For wind-solar-storage stations, the transactional electricity volume Δt_a for day *t* and period *a* is determined as follows:

$$\begin{aligned} &\Delta t_a = Q_{ta} \cdot \left(C - Q_{poverty_ta}\right) / \\ &\max\{C - Q_{poverty_ta}\, Q_{ta} + Q_{12_ta} \\ &+ Q_{parity_ta} + Q_{wu_ta} + Q_{normal_ta}\} \end{aligned} \tag{9}$$

Where:Q_{ta}, Q_{12_ta}, Q_{parity_ta}, Q_{wu_ta}, Q_{normal_ta} represent the power generation of the wind-solar-storage station, 12 subsidized wind farms, parity wind farms, Wudenshan wind farm, and normal wind farms on day *t* and period *a*, respectively. *C* denotes the capacity limit of the trading channel, and $Q_{poverty_ta}$ represents the power generation of the photovoltaic poverty alleviation station on day *t* and period *a*.

(2) Some Units Participate in the Tertiary Market

1) Wind-Solar-Storage Station Participates in the Tertiary Market.

For day t and period, a, the transactional electricity volume Δt_a of the wind-solar-storage station is determined as follows [27]:

$$Q_{ta} + Q'_{ta} \le C - Q_{poverty_ta}, \Delta t_a = Q_{ta} \tag{10}$$

$$\begin{aligned} &Q_{ta} + Q'_{ta} C - Q_{poverty_ta}, \Delta t_a = \\ &Q_{ta}\left(C - Q_{poverty_\mathrm{ia}}\right) / \left(Q_{ta} + Q'_{ta}\right) \end{aligned} \tag{11}$$

Where Q'_{ta} represents the power generation of other units participating in the tertiary market on day *t* and period *a*.

(1) Wind-Solar-Storage Station Participates in the Tertiary Market

(2) Wind-Solar-Storage Station Does Not Participate in the Tertiary Market

$$Q'_{ta} C - Q_{poverty_ta}, \Delta t_a' = 0 \tag{12}$$

$$Q'_{ta} \le C - Q_{poverty_ta},$$
$$\Delta t_a' = Q_{ta}\left(C - Q_{poverty_ta} - Q'_{ta}\right) \tag{13}$$
$$/\max\left\{C - Q_{poverty_ta} - Q'_{ta}, Q''_{ta} + Q_{ta}\right\}$$

Where $\Delta t_a'$ is the trading volume in the fourth-level market for the wind-solar-storage station on day *t* and period *a*. Priority is given to other market members participating in the tertiary market. When the channel capacity reaches the limit, the wind-solar-storage station, being of lower priority, cannot trade any electricity in this period, resulting in a trading volume of 0. Q''_{ta} represents the power generation of other units participating in the fourth-level market on day *t* and period *a*.

The actual grid constraints include:

$$P_t \le P_{\max}, V_{\min} \le V_i \le V_{\max}$$
$$f_{\min} \le f \le f_{\max}, P_l = P_r + P_d \tag{14}$$

Where P_{loss} is Power loss, P_t is current transmission line resistance, P_{max} is actual transmitted power, maximum grid capacity, V_i is node voltage, V_{max} is maximum value, V_{min} is minimum value, f isgrid frequency, f_{min} is minimum frequency, f_{max} is maximum frequency, P_l is total load demand, P_r is renewable energy generation, P_d is dispatchable generation.

(3) No Units Participate in the Tertiary Market

In this case, all members have equal priority, and the method for determining the trading volume of the wind-solar-storage unit on day *t* and period *a* follows the same rules as in (1).

To systematically assess the impact of discount rate settings on economic indicators and project ranking results, this study incorporates a discount rate sensitivity analysis module into the baseline economic evaluation framework. Specifically, two financing scenarios are designed to reflect typical differences in capital cost structures: (i) a green bond financing scenario representing policy-supported low-cost capital, with a relatively low discount rate (e.g., 4%); and (ii) a conventional commercial loan scenario representing market-based financing, with a higher discount rate (e.g., 8%). Under each discount rate setting, key economic indicators such as net present value (NPV) and benefit-cost ratio (B/C) are calculated for all projects. A weighted scoring method is then applied to compare project rankings, thereby revealing the sensitivity of evaluation results to discount rate variations and highlighting the underlying influence of financing structure on project performance.

The ancillary service revenue factor ∂_s is defined as the ratio of revenue from ancillary services to the total project revenue:

$$\partial_s = R_{AS}/R_t \tag{15}$$

This factor quantifies the economic contribution of grid services such as frequency regulation, voltage support, and spinning reserve.

The policy support coefficient κ_p quantifies the proportional increase in project revenue attributable to policy incentives, such as subsidies, tax credits, or feed-in tariffs. It is defined as:

$$\kappa_p = R_p/R_b \tag{16}$$

Where: $R_b$is Baseline revenue without policy support; R_p is Revenue from subsidies, incentives, or policy-driven mechanisms

2.2 Environmental Indicators

Charging and Discharging Power Constraints The new energy stations have demonstrated significant environmental benefits in green power trading projects, particularly in terms of zero emissions, which are clearly superior to traditional coal-fired power generation. Wind power projects not only meet electricity demand but also effectively reduce reliance on coal-fired power. By comparing with coal-fired power plants of similar scale, the tangible contribution of green power projects in reducing emissions can be clearly demonstrated.[28, 29]

The current model assesses environmental performance primarily through carbon emission (CO_2) reduction. While important, carbon emissions capture only part of the full life-cycle impact of renewable energy. To address this limitation, we propose integrating Life Cycle Assessment (LCA) into the model, using standardized frameworks such as ReCiPe, TRACI, or ILCD to evaluate multiple environmental factors across wind and solar projects. Key additions include land use (measured as area per installed capacity, m^2/MW), rare earth material use (based on component bills of materials), and a new "Critical Raw Material Scarcity Index" as a sub-indicator in the environmental dimension.

$$LUI = A_t/P_i \tag{17}$$

$$WUI = \frac{V_w}{E_o} \cdot WF_s \tag{18}$$

$$RSS = \sum_i m_i \cdot cF_i \tag{19}$$

Where, Eq. (15) represents the land use intensity per unit of installed capacity; Eq. (18) represents the water resource consumption factor; Eq. (19) denotes the Critical Raw Material Scarcity Index. A_t is the total land area used during the project's life cycle (m^2); P_i is the installed capacity of the power station (MW). V_w is the total water consumption over the life cycle (m^3); E_o is the total electricity generation during the life cycle (MWh); WF is the regional water scarcity weighting factor; m_i is the amount of the i-th critical raw material used over the product's life cycle (kg); cF_i is the scarcity impact factor of that material.

Considering electricity price and carbon price as two random variables, the correlation coefficient matrix $\rho$is expressed as:

$$\rho = \begin{bmatrix} 1 & \rho_{ec} \\ \rho_{ce} & 1 \end{bmatrix} \tag{20}$$

$$V_4 = \frac{V_3 R_6 R_7 + 3.3 R_5 R_6}{R_5 R_6 + R_5 R_7 + R_6 R_7} \tag{21}$$

Where, ρ_{ec} and ρ_{ce} represent the correlation coefficients of electricity to carbon and carbon to electricity, respectively. According to the mathematical property of the correlation coefficient matrix, we have ρ_{ec}=ρ_{ce}.

Let the correlation coefficient matrix be denoted as ρ, and the sample matrix Y is characterized by variances ρ_i, where i=1or2:ρ_1 is the variance of electricity prices,ρ_2 is the variance of carbon prices.

Therefore, the sampled data Y_t for the correlation between electricity price and carbon price on day t is given by:

$$Y_t = diag\left(\omega_i\right) \cdot Z_{pt} + \lambda_{ec,t} \tag{22}$$

Where, **diag** denotes the diagonalization of the sample variance matrix for electricity and carbon prices. $\lambda_{ec,t}$ represents a 2×N matrix of average values for electricity and carbon prices on day *t*, composed of the daily mean electricity price $\lambda_{ec,t}$ and carbon price $\lambda_{ec,t}$. Based on this, during the optimization period S (set as 31 days), *N* samples are drawn per day to generate correlated sample matrices of electricity and carbon prices, denoted as Yt=(Ye,t,Yc,t) for *t*=1,2,…,N. The time unit for the optimization period is days.

The following are the environmental evaluation indicators:

$$C_G = \sum_{i=1}^{n} Q_{Gi} / \sum_{j=1}^{m} Q_j \tag{23}$$

$$G_{CO2} = Q_{CO2} / Q_E \tag{24}$$

Where Formula (23) is Green Power Trading Ratio; Q_{Gi} is the trading electricity volume of the i-th green power unit, with a total of n green power units;Q_j is the trading electricity volume of the j-th unit, with a total of m units. Formula (24) is CO_2 Emission Intensity per Unit of Electricity; Q_{CO2} is the total CO_2 emissions in the region;Q_E is the total market transaction electricity volume.

Due to the volatility and intermittency of wind and photovoltaic (PV) units, there is often a mismatch between their output curves and the system load curve. To promote the integration of clean wind and solar power, energy storage systems (ESS) can effectively store the excess electricity generated during periods of high output and low demand. When the output from wind and PV units is low and the system demand is high, the stored energy can be supplied to the grid. This not only enhances energy utilization efficiency but also improves the safety and stability of power systems with high penetration of renewable energy sources such as wind and solar. The constraints of the energy storage system are as follows:

$$0 \le \alpha_{i,t}^{\mathrm{ESS}} + \beta_{i,t}^{\mathrm{ESS}} \le 1,\ \alpha_{i,t}^{\mathrm{ESS}},\ \beta_{i,t}^{\mathrm{ESS}} \in \{0,1\},\ \forall i,t \tag{25}$$

$$0 \le P_{i,t,\alpha}^{\mathrm{ESS}} \le \alpha_{i,t}^{\mathrm{ESS}} P_{i,\alpha,max}^{\mathrm{ESS}},\ \forall i,t \tag{26}$$

$$0 \le P_{i,t,\beta}^{\mathrm{ESS}} \le \beta_{i,t}^{\mathrm{ESS}} P_{i,\beta,max}^{\mathrm{ESS}},\ \forall i,t \tag{27}$$

$$P_{i,t}^{\mathrm{ESS,E}} = P_{i,\beta,t}^{\mathrm{ESS}} - P_{i,\alpha,t}^{\mathrm{ESS}},\ \forall i,t \tag{28}$$

$$P_{i,t}^{\mathrm{ESS,E}} + P_{i,t}^{\mathrm{ESS,RES}} + P_{i,t}^{\mathrm{ESS,RAMPU}} \le P_{i,\beta,max}^{\mathrm{ESS}},\ \forall i,t \tag{29}$$

$$P_{i,t}^{\mathrm{ESS,E}} - P_{i,t}^{\mathrm{ESS,RAMPD}} \ge -P_{i,\alpha,max}^{\mathrm{ESS}},\ \forall i,t \tag{30}$$

$$P_{i,t}^{\mathrm{ESS,E}} h^{\mathrm{E}} + P_{i,t}^{\mathrm{ESS,RES}} h^{\mathrm{RES}} + P_{i,t}^{\mathrm{ESS,RAMPU}} h^{\mathrm{RAMP}} \le E_{i,t}^{\mathrm{ESS}},\ \forall i,t \tag{31}$$

$$E_{i,t}^{\mathrm{ESS}} - P_{i,t}^{\mathrm{ESS,E}} h^{\mathrm{E}} + P_{i,t}^{\mathrm{ESS,RAMPD}} h^{\mathrm{RAMP}} \le SOC_{i,\max}^{\mathrm{ESS}} C_i^{\mathrm{ESS}},\ \forall i,t \tag{32}$$

$$E_{i,t}^{\mathrm{ESS}} = E_{i,t-1}^{\mathrm{ESS}} + \left(P_{i,\alpha,t}^{\mathrm{ESS}} \eta_{i,\alpha}^{\mathrm{ESS}} - P_{i,\beta,t}^{\mathrm{ESS}} / \eta_{i,\beta}^{\mathrm{ESS}}\right) h,\ \forall i,t \tag{33}$$

$$SOC_{i,\min}^{\mathrm{ESS}} \le E_{i,t}^{\mathrm{ESS}} / C_i^{\mathrm{ESS}} \le SOC_{i,\max}^{\mathrm{ESS}},\ \forall i,t \tag{34}$$

$$E_{i,0}^{\mathrm{ESS}} = E_{i,T}^{\mathrm{ESS}},\ \forall i \tag{35}$$

To address the differences in output characteristics, capacity factors, and dispatch responsiveness among wind, PV, and wind–solar–storage power stations, this study introduces a type-specific adjustment mechanism within a unified

evaluation framework to enhance fairness and comparability. For hybrid stations, two dedicated indicators—dispatch response efficiency and storage utilization efficiency—are included to reflect their capabilities in fast regulation and peak shaving. This extension improves the model's ability to capture complex operational behaviors, mitigates systemic bias across heterogeneous technologies, and ensures the comparability and discriminative power of the evaluation results.

This study presents a preliminary degradation modeling approach. Specifically, based on typical performance characteristics of lithium-ion batteries, the metric of Full Equivalent Cycles (FEC) was adopted to quantify the cumulative fatigue of the battery system. A simplified rainflow counting algorithm was employed to statistically analyze charge-discharge cycles under simulated operational conditions.

Based on this analysis, degradation-related costs—including capacity loss, replacement expenses, and operation and maintenance—were converted into an equivalent discounted cost per unit of delivered energy and incorporated into the lifecycle economic evaluation framework. Simulation results indicate that, after accounting for degradation costs, the net present value (NPV) of some storage-dominant projects exhibited a slight decline, leading to minor adjustments in their economic ranking. However, the overall impact remained within an acceptable range, demonstrating the flexibility and robustness of the proposed evaluation model.

we conducted an extended analysis to assess the techno-economic and environmental impacts of coupling renewable energy projects with green hydrogen electrolyzer systems. From an economic perspective, green hydrogen loads can serve as a flexible consumption pathway for renewable electricity, thereby reducing curtailment and enhancing the overall utilization rate of generated power. In addition, hydrogen products represent a supplementary revenue stream that may improve project profitability under certain market conditions.

From an environmental standpoint, green hydrogen—being a zero-carbon energy carrier—contributes significantly to carbon reduction. Its co-location with renewable energy infrastructure enhances the project's environmental performance metrics. To evaluate the adaptability of our model under this scenario, a green hydrogen electrolyzer load module was incorporated as a dispatchable flexible demand into the power system simulation. Key operational characteristics such as dynamic responsiveness, electrolyzer efficiency, and hydrogen market price fluctuations were explicitly considered.

Simulation results indicate that under the green hydrogen coupling scenario, several projects previously constrained by grid integration limitations showed an improvement in their overall scores, particularly in the environmental dimension. These findings demonstrate the flexibility of the proposed framework and its potential for future expansion to more integrated "generation–load–storage–hydrogen" evaluation pathways.

2.3 Social Indicators

While large-scale project evaluations have traditionally focused on capital investment, cost, and output, growing emphasis on sustainability highlights the need to consider demographic shifts and employment impacts. Modern projects must therefore integrate social benefit assessments to inform policymaking and align with development goals. Given the qualitative and complex nature of social impacts, a multidimensional framework—spanning employment, energy structure, and quality of life—is essential [30].

Chinese wind and solar firms have become global green energy leaders, generating economic returns and substantial employment across construction, operations, and supply chains. Digitalization and automation have further raised labor quality by increasing skilled job demand. These projects also drive regional development by enhancing infrastructure and environmental quality. As coal remains dominant, promoting renewable participation in green power trading and strengthening social benefit evaluation are key to optimizing the energy mix and advancing sustainable development.

$$I_{MP} = \sum_{i=1}^{m} Q_{pi} / \sum_{j=1}^{n} Q_{tj} \tag{36}$$

Where Formula (36) is Market Concentration; Q_{pi} is The capacity of the i-th power generation company or user that meets the market access conditions Q_{tj} is the capacity of the j-th power generation company or user. $\eta = F/C_{act}$ is Market Competition Level. η is the market competition index;F is the total electricity purchase cost;$C_{act}$is the actual operating cost.

To test the robustness of social benefit evaluation results, a Monte Carlo simulation is applied to perturb sub-indicator weights. Starting from the expert-defined baseline weights, a±20% variation range is set for each indicator. In each iteration, new weights are randomly generated within this range, normalized to sum to 1, and used to recalculate benefit scores and rankings.

To incorporate the effects of negative externalities such as visual impact and noise, the social dimension scoring model was revised by introducing a penalty term. The modified social score is calculated as follows:

$$S_s = S_b - \gamma_{NE}\left(S_v + S_n\right) \tag{37}$$

Where γ_{NE} is Negative Externality Coefficient; S_v is Visual Impact Score; S_n is Noise Impact Score

2.4 Extreme Weather Uncertainty Model

To further account for the revenue uncertainty of variable renewable energy sources under extreme weather conditions, this study introduces a disturbance-factor-based extreme weather scenario modeling framework. The approach aims to simulate the curtailment effects of adverse weather events—such as prolonged cold spells, extended windless periods, and sandstorms—on wind and solar power generation. Based on this, the model assesses the corresponding impacts on economic and social benefit indicators.

(1) Definition of Disturbance Factor

Assuming that under normal weather conditions, the annual potential power generation of a given project is G_0 (unit: MWh), the effective power generation under extreme weather conditions is expressed as:

$$G = G_0 \cdot (1 - \omega) \tag{38}$$

Where $\omega \in [0,0.5]$ is the disturbance factor, representing the proportion of power generation loss caused by extreme weather conditions. Based on historical weather data and empirical findings from the literature, this study defines three levels of scenario-based disturbances: mild ($\omega = 0.2$), severe ($\omega = 0.3$), and extreme ($\omega = 0.4$).

(2) Adjustment of Economic Indicators

Considering that reduced power generation directly leads to an increase in the unit electricity price, this study adopts a modified Levelized Cost of Electricity (LCOE) calculation model.

$$LCOE = C/\left(G \cdot 1000\right) \tag{39}$$

Where CCC is the total investment cost of the project (in RMB), and GGG is the power output after disturbance, converted to kilowatt-hours (kWh).

(3) Adjustment of Social Benefit Index

Given that unstable energy supply can negatively affect social indicators such as employment and public acceptance, this study introduces a linearly reduced correction value for social benefits based on the degree of revenue decline.

$$S = S_0 \cdot (1 - K \cdot \omega) \tag{40}$$

Where S_0 is the social benefit index in a normal year, and k is the social sensitivity coefficient (set to 1.0 in this study), reflecting the responsiveness of the social dimension to power generation disturbances.

3 Evaluation System

3.1 Establish a Hierarchy

By developing a hierarchical analytical framework that includes economic, environmental, and social indicators (as shown in Fig. 1), this study systematically explores the performance and potential of new energy stations in green power trading projects. This approach provides valuable

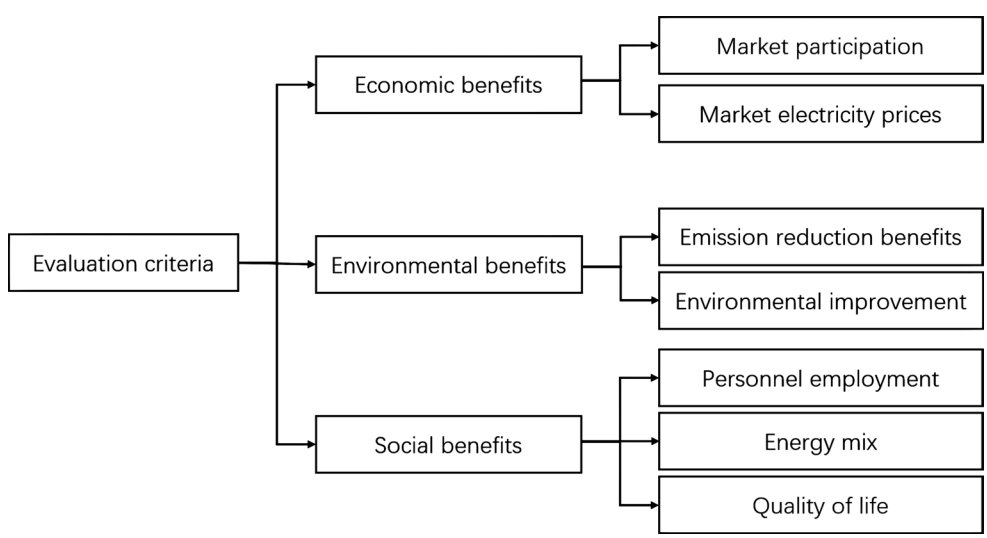


Fig. 1 Analytic hierarchy framework

insights and recommendations for research and practice in related fields.

3.2 AHP-Entropy Method

In constructing the decision analysis framework, we employ the 1–9 scale method to conduct detailed pairwise comparisons of the indicators, thereby creating a rigorous comparison judgment matrix. For indicators within the same objective and evaluation criteria, we systematically compare their relative importance and assign corresponding weight values based on expert insights.

Specifically, experts, drawing on their experience and professional knowledge, subjectively assess the relative importance of any two indicators within each level of the hierarchy. These subjective judgments are converted into numerical values using the standardized 1–9 scale method. This quantification process allows us to build a clear judgment matrix from these numerical values.

The weight calculation model is constructed as follows:

$$\mathrm{M} = \begin{bmatrix} A_{11}\ \mathrm{A}_{12}\ \cdots\ \mathrm{A}_{1m} \\ \mathrm{A}_{21}\ \mathrm{A}_{22} \cdots \mathrm{A}_{2m} \\ \cdots\cdots\cdots\cdots \\ A_{m1}\ \mathrm{A}_{m2} \cdots \mathrm{A}_{mm} \end{bmatrix} \tag{41}$$

$$f_i = \sum_{j=1}^{i} \left(A_{ij} / \sum_{i=1}^{m} A_{ij} \right) (i, j = 1, 2, \ldots, m) \tag{42}$$

Where M represents the pairwise comparison matrix constructed using the 1–9 scale method for comparing indicators. x_{ij} denotes the normalized pairwise comparison matrix. f_i signifies the weight for each indicator.

After obtaining the weights, avoid the inconsistency errors in logical judgment that may be caused by human factors, and conduct consistency testing:

$$CR = \left\{ \left(\sum_{i=1}^{m} \frac{A\left(f_i / \sum\limits_{i=1}^{m} f_i\right)}{m\left(f_i / \sum\limits_{i=1}^{m} f_i\right)_i} - m \right) / (m-1) \right\} \tag{43}$$
$$/RI\,(i = 1, 2, \ldots, m)$$

Where $\overline{f_i}$ represents the weight vector $\lambda_{\max}$ is the maximum eigenvalue of the matrix. CI is the Consistency Index.CR is the Consistency Ratio.RI is the Random Consistency Index (as shown in Table 1).

When CR0.1 the consistency is considered satisfactory, and the judgment matrix passes the consistency test. Otherwise, the judgment matrix should be reconstructed or adjusted until it meets the consistency criteria.

The entropy method, an objective weighting approach based on information entropy, determines indicator importance by measuring system uncertainty, thus improving the scientific and unbiased nature of evaluations. To assess potential correlation or redundancy among secondary indicators in the comprehensive benefit model—and avoid "double counting"—Principal Component Analysis (PCA) is applied for dimensionality reduction and correlation analysis.

Assume the comprehensive evaluation indicator matrix is $X = [x_{ij}] \in R^{m \times n}$, where m denotes the number of project samples, and n is the total number of standardized secondary indicators. The matrix X is processed as follows:

Standardization:

All indicators are standardized using zero-mean and unit-variance normalization to construct the standardized matrix Z:

$$Z_{ij} = \frac{x_{ij} - \mu_j}{\sigma_j} \tag{44}$$

Where $\mu_j$and $\sigma_j$represent the sample mean and standard deviation of the j-th indicator, respectively.

Covariance Matrix Calculation:The covariance matrix Σ of the standardized indicators is constructed as:

$$\sum = \frac{1}{m-1} Z^T Z \tag{45}$$

Eigenvalue Decomposition and Principal Component Extraction:The covariance matrix Σ is then subjected to eigenvalue decomposition to obtain eigenvectors V_k and corresponding eigenvalues λ_k. The principal components corresponding to the top ppp eigenvalues are selected as the new uncorrelated variables:

$$Y = Z \cdot V_p \tag{46}$$

Where Vp=[v1,v2, … ,vp] is the matrix composed of the top p eigenvectors, such that the cumulative contribution rate satisfies:

$$\sum_{k=1}^{p} \lambda_k / \sum_{k=1}^{n} \lambda_k \geq \tau\,(\tau = 0.85 \sim 0.95) \tag{47}$$

Table 1 Table of RI values

1	2	3	4	5	6	7	8	9
0.00	0.00	0.58	0.90	1.12	1.24	1.32	1.41	1.45

Information entropy reflects the level of uncertainty in a system: the higher the entropy, the greater the uncertainty within the system; conversely, the lower the entropy, the higher the certainty of the information. In the entropy method, the entropy value of an indicator is inversely proportional to its assigned weight—i.e., the higher the entropy value, the smaller the importance weight of the corresponding indicator; conversely, the lower the entropy value, the greater the weight. This characteristic ensures that the evaluation system can objectively reflect the relative contributions of each indicator to the overall evaluation.The steps of the entropy method are as follows:

(1) Data Standardization

First, the raw data are standardized to eliminate the influence of different units and scales across indicators. Common standardization methods include range normalization and Z-score normalization.Range Normalization: The data are proportionally transformed into the [0, 1] range.

$$y'_{ij} = \left(y_{ij} - y_{i\,\min}\right) / \left(y_{i\,\max} - y_{i\,\min}\right) \tag{48}$$

where y_{ij} represents the raw data of the indicator for the j sample, and y_{imin} and y_{imax} are the minimum and maximum values of the indicator, respectively.

(2) Calculation of the Entropy Value for Each Indicator

For each indicator, its entropy value Y_i is calculated. The entropy value reflects the degree of dispersion in the data for that indicator. The calculation steps are as follows:

$$Y_i = -\frac{1}{\ln(n)} \sum_{j=1}^{n} \left(y'_{ij} / \sum_{j=1}^{n} y'_{ij} \right) \ln \left(y'_{ij} / \sum_{j=1}^{n} y'_{ij} \right) \tag{49}$$

where n is the number of samples, and y_{ij} is the normalized value of the i-th indicator for the j sample.

(3) Calculation of Indicator Weights

Based on the entropy values, the weight ω_i for each indicator is calculated. The greater the entropy value, the smaller the weight of the corresponding indicator. The calculation steps are as follows:

$$\omega_i = (1 - Y_i) / \sum_{i=1}^{m} (1 - Y_i) \tag{50}$$

where m is the number of indicators.

The combination weighting method integrates multiple techniques—typically both subjective (e.g., AHP) and objective (e.g., entropy or CRITIC methods)—to improve the scientific and balanced allocation of weights. It involves four steps: (1) selecting representative methods to capture expert judgment and data characteristics; (2) calculating weights separately using each method; (3) combining them via arithmetic averaging, weighted averaging, or optimization; and (4) normalizing the final weights to ensure they sum to 1. By fusing diverse information sources, this method enhances weight reliability and decision-making rationality.

Through combination weighting, the combined weights based on the AHP-entropy method are further determined:

$$W = \beta \cdot W_{AHP} + (1 - \beta)\, W_e \tag{51}$$

Where β represents the proportion of the indicator weights determined by the AHP method in the combined weights, and $1 - \beta$ represents the proportion of the indicator weights determined by the entropy method in the combined weights. In this study, based on discussions and analyses from multiple experts, β is set to 0.3. Therefore, the combined weights based on the AHP-entropy method are:

In this study, the parameter β=0.3 is initially set based on two main considerations:First, the expert evaluation system in this field is supported by substantial domain experience;Second, the entropy weight method provides an objective complement, particularly when dealing with high-dimensional data.The weight setting of 0.3 demonstrated good explanatory power and ranking stability during preliminary research.To further validate the rationality of the chosen β value and assess the model's robustness, a sensitivity analysis is conducted. The value of β is varied within the range of 0.1 to 0.9 in increments of 0.1, resulting in 9 different combined weighting schemes. The comprehensive benefit scores and ranking results under each scheme are then compared to evaluate consistency and sensitivity.

3.3 Comprehensive Evaluation Model

The comprehensive evaluation method integrates multiple factors to provide a thorough and effective assessment. It employs fuzzy set theory to manage inherent ambiguity, transcending precise mathematical logic to deeply describe various indicators. As a method that combines qualitative and quantitative analysis, fuzzy comprehensive evaluation is widely used in fields such as engineering technology, economic management, and social life. Therefore, applying fuzzy mathematics to address comprehensive evaluation issues is very appropriate. The rules for applying fuzzy comprehensive evaluation are as follows [31]:

Selection of Factors: Construct a comprehensive benefit evaluation index system. Formula includes 7 specific indicators, which assess and describe the participation of new energy stations in green power trading projects from the perspectives of economic, social, and environmental factors.

$$U = \{u_1, u_2, u_3, u_4, u_5, u_6, u_7\} \tag{52}$$

Selection of Evaluation Linguistic Terms: Formula specifies the number of fuzzy evaluation linguistic terms for the project, which are: Excellent, Good, Average, Poor, and Very Poor. These terms constitute the linguistic term set for the comprehensive benefit evaluation of new energy stations participating in green power trading projects.

$$V = \{v_1, v_2, v_3, v_4, v_5\} \tag{53}$$

Constructing the Fuzzy Matrix: For each factor u_i in the factor set U, perform a separate evaluation to calculate the degree of membership of each factor to the linguistic terms set v_i.This results in obtaining the linguistic term set for each factor u_i,which forms the fuzzy relationship matrix.

$$R = \begin{bmatrix} \lambda_{11}\lambda_{12}\cdots\lambda_{1n} \\ \lambda_{21}\lambda_{22}\cdots\lambda_{2n} \\ \cdots\cdots\ddots\cdots \\ \lambda_{m1}\lambda_{m2}\cdots\lambda_{mn} \end{bmatrix} \tag{54}$$

Factor Weight Vector W: Based on the final weights of each indicator, construct the weight vector W for the factor set. The weight vector represents the importance of each factor in the comprehensive evaluation [32–35].

Calculate Results: Using fuzzy statistical methods, invite N experts to score all indicators and calculate the degree of membership for each indicator. Based on these single-factor evaluation results, perform a comprehensive fuzzy evaluation using Formula with weighted average and fuzzy operators for multiplication and addition.

$$Z = W\left(\bullet, \oplus\right) R \tag{55}$$

To classify the fuzzy scores of each evaluation object into qualitative categories, this study constructs a five-level evaluation system using triangular membership functions. The levels are defined as: “Excellent”, “Good”, “Average”, “Poor”, and “Very Poor”, with their membership functions defined over the score interval [0,1]. The peak (center) points of the triangular functions are set as follows: $\mu_{\text{Excellent}}(x) \to 0.90,\ \mu_{\text{Good}}(x) \to 0.75,$ $\mu_{\text{Average}}(x) \to 0.60,\ \mu_{\text{Poor}}(x) \to 0.45,$ $\mu_{\text{VeryPoor}}(x) \to 0.30$

Triangular Membership Function Example (for “Good”):For the “Good” level, the triangular membership function $\mu_{\text{Good}}(x)$ can be defined as:

$$\mu_{Good}(x) = \begin{cases} 0, x\ \langle 0.65, x\rangle\ 0.85 \\ \frac{x-0.65}{0.75-0.65}, 0.65 \le x \le 0.75 \\ \frac{0.85-x}{0.85-0.75}, 0.75 < x \le 0.85 \end{cases} \tag{56}$$

The complete parameters for the five evaluation levels are shown in Table 2.

The general form of all triangular membership functions is:

$$\mu(x) = \begin{cases} 0, x \le a, x \ge b \\ \frac{x-a}{c-a}, a < x \le c \\ \frac{b-x}{b-c}, c < x < b \end{cases} \tag{57}$$

The model’s overlapping evaluation levels reflect the real-world fuzziness of linguistic assessments, ensuring both differentiation and flexibility. To address the subjectivity and ambiguity inherent in fuzzy AHP, a±10% perturbation is applied to each triangular fuzzy number, and Monte Carlo simulation is used to generate varied judgment matrices. Recalculated rankings and stability metrics—such as top-tier consistency and score standard deviations—are used to construct fuzzy confidence intervals, enhancing result credibility.

The model demonstrates strong robustness and adaptability, making it applicable across both theoretical and practical contexts. Its ability to integrate diverse factors under varying conditions provides a solid foundation for future research and real-world deployment.

4 Case Studies

4.1 Description

This study evaluates the comprehensive performance of six typical new energy stations in northern Hebei—including poverty alleviation PV, standard PV, subsidized and parity

Table 2 Parameters for the five evaluation levels

Level	Membership function center	Left boundary a	Right boundary	Mathematical expression
Excellent	0.3	0.2	0.4	Triangular Function
Good	0.45	0.35	0.55	Triangular Function
Average	0.6	0.5	0.7	Triangular Function
Poor	0.75	0.65	0.85	Triangular Function
Very Poor	0.9	0.8	1	Triangular Function

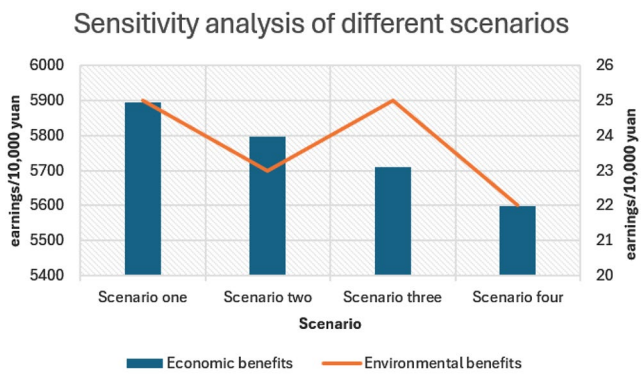


Fig. 2 Sensitivity analysis of different scenarios

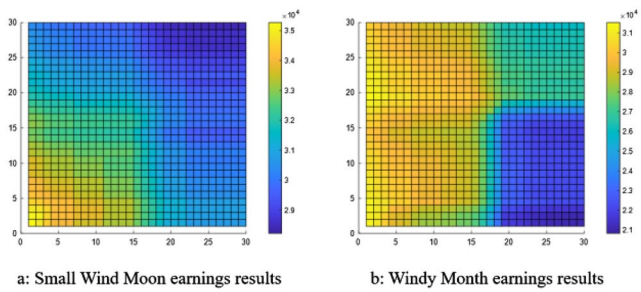


Fig. 3 Earnings results

wind farms, Wudengshan wind, and wind–solar–storage hybrids—based on one year of operational data within the context of green power trading.

A three-dimensional evaluation framework is developed, covering economic, environmental, and social aspects. The economic dimension uses a nonlinear optimization model incorporating real-time electricity and carbon prices under transmission constraints. The environmental dimension expands beyond carbon reduction to include life cycle indicators such as land use, water consumption, and critical material scarcity. The social dimension introduces a multi-indicator system (e.g., employment, infrastructure), with Monte Carlo simulations applied to test weight robustness via ± 20% perturbations.

To benchmark the model's predictive accuracy and ranking stability, statistical metrics (RMSE, MAE, MAPE, R^2) are calculated, and paired-sample t-tests are used to assess the significance of ranking differences ($\alpha = 0.05$). Additionally, various extreme weather scenarios (e.g., low wind, low irradiance, dust storms) are simulated to evaluate the stabilizing effect of energy storage.

Results confirm the framework's ability to capture the multi-dimensional value of new energy stations and its utility in supporting strategic decisions.

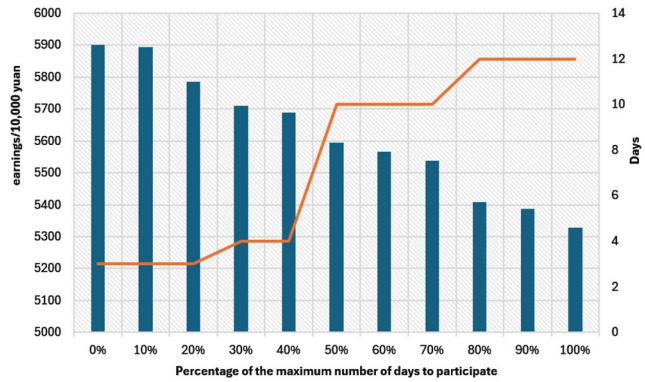


Fig. 4 Maximum benefits

4.2 Revenue from New Energy Stations

Figure 2 shows the impact of different electricity prices on each indicator, conducting a sensitivity analysis for different scenarios.

This study evaluates the comprehensive benefits of a new energy station across economic, environmental, and social dimensions. Compared to traditional single-method evaluations, the proposed method demonstrates superior accuracy, reliability, and comprehensiveness.

Figure 3 compares optimal participation days in the third-tier market under low and high wind output scenarios. In the Small Wind Month, when others rarely participate (less than 1 day), the optimal strategy is for the station to avoid participation. As others participate more (e.g., 2 days), the optimal participation shifts to 3–19 days, maximizing revenue.

Using October data from a wind-solar-storage station, Fig. 4 shows that when other stations declare over 80% of their maximum allowed days, the optimal strategy for the unit is 13 participation days. As others' participation increases, the station's revenue first peaks but later declines due to transmission congestion, which limits electricity sales.

These results validate the practical value of the proposed model, showing its ability to adapt bidding strategies based on market behavior and improve economic outcomes for new energy stations under dynamic trading conditions.

4.3 Metric Weights

Based on the quantitative and qualitative analysis of the case study results, the weight results are as shown in Tables 3, 4, 5, 6, and 7.

4.4 Comprehensive Benefits

Figure 5 illustrates the indicator weights and comprehensive evaluation results for renewable energy stations

Table 3 Target layer weights

Comprehensive benefits	Economic	Social	Environmental	weight
Economic	1	7	5	0.731
Social	1/7	1	3	0.08
Environmental	1/5	1/3	1	0.189
C.R. = 0.0516 < 0.1，Pass the consistency test				

Table 4 Weights of economic indicators

Economic indicators	Market participation	Market electricity prices	weight
Market participation	1	3	0.75
Market prices	1/3	1	0.25
C.R. = 0 < 0.1，Pass the consistency test			

Table 5 Environmental indicator weights

Environmental indicators	Carbon emissions	environmental pollution	weight
Carbon emissions	1	1/4	0.2
environmental pollution	4	1	0.8
C.R. = 0.0000 < 0.1，Pass the consistency test			

Table 6 Weights of social indicators

Social indicators	Personnel employment	Energy mix	quality of life	weight
Personnel employment	1	3	5	0.648
Energy mix	1/3	1	2	0.230
quality of life	1/5	1/2	1	0.122
C.R. < 0.1，Pass the consistency test				

Table 7 Table of total weights

Target layer	Criterion layer
Economic indicators (0.731)	Market participation (0.5483)
	Market electricity prices (0.1827)
Environmental indicators (0.08)	Carbon emissions (0.016)
	environmental pollution (0.064)
Social indicators (0.189)	Personnel employment (0.1225)
	Energy mix (0.0435)
	quality of life (0.023)

participating in green electricity trading. Each indicator contributes differently, with its assigned weight directly influencing the overall evaluation outcome and reflecting its relative importance in decision-making.

Figure 6 presents the results of a single-benefit assessment. Among all factors, economic benefits hold the highest weight and serve as the primary criterion. Although environmental protection, energy efficiency, technological maturity, and policy support carry smaller weights, they still play a meaningful role in shaping the final evaluation.

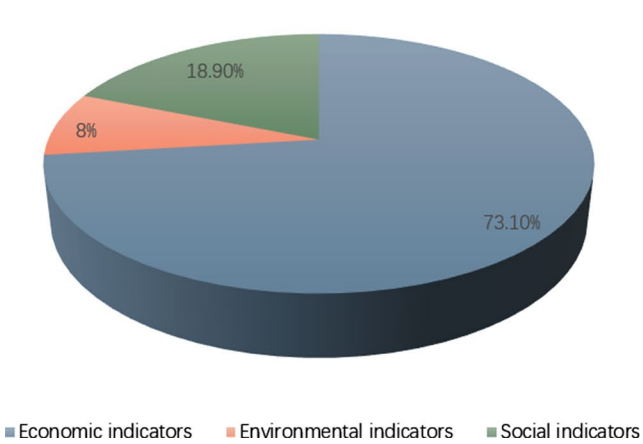
From the statistical comparison results in Table 8, the proposed model outperforms the benchmark methods in terms of RMSE, MAE, and R^2, demonstrating lower error margins and higher fitting accuracy. Moreover, a paired-sample t-test conducted on the ranking results of the model outputs indicates that the difference between the proposed model and the baseline model is statistically significant at the 95% confidence level ($p = 0.002$), further confirming its superiority in multidimensional benefit evaluation. These findings suggest that the newly introduced multidimensional weighting structure and comprehensive benefit assessment framework significantly enhance prediction accuracy and ranking differentiation while maintaining theoretical soundness.

The results of the β sensitivity analysis show in Table 9 that when β varies within the range [0.1,0.9], the comprehensive scores of each project exhibit only minor fluctuations, while the overall ranking structure remains stable. Specifically, the top five projects shift by no more than one position across different β values, and the ranking consistency, measured by Kendall's coefficient, remains above 0.95, indicating strong robustness of the model with respect to the β setting.Additionally, for a few borderline projects, rankings slightly improve when β is larger (i.e., when AHP weights dominate), suggesting that subjective preferences may amplify sensitivity to specific indicators in certain scenarios. Overall, the choice of $\beta = 0.3$ demonstrates a good balance between subjective and objective weighting, with strong stability and adaptability.

This study further employed Monte Carlo perturbation simulations to conduct sensitivity testing and construct confidence intervals for the fuzzy AHP results. Across 5,000 simulation iterations, the standard deviation of each project's score remained below 0.02, and the top three rankings were consistent in 95% of the simulations. This approach effectively addresses a key limitation of traditional fuzzy AHP—namely, its tendency to produce deterministic outputs while neglecting underlying uncertainty.

5 Conclusion

In conclusion, the proposed comprehensive benefit evaluation system demonstrates strong scientific validity, structural balance, and practical applicability. By integrating economic, environmental, and social dimensions within a fuzzy-entropy framework, the model improves the rationality of indicator weighting and the credibility of evaluation outcomes. It accounts for real-world factors such as transmission constraints, price volatility, storage limitations, and extreme weather, enhancing robustness and adaptability. Empirical results from six types of new energy stations show that 75.32% of projects are rated "excellent" under this model, compared to 69.42% under traditional methods—a statistically significant improvement. The framework effectively supports optimal bidding strategies,



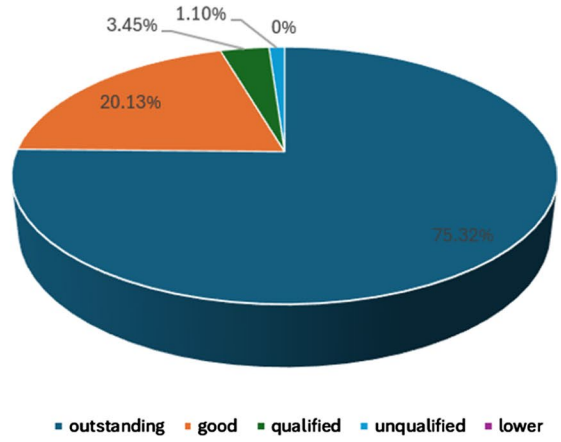


Fig. 5 Comprehensive benefit evaluation results

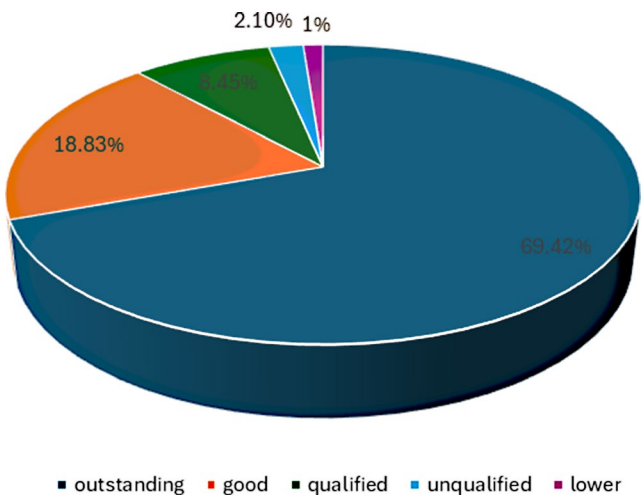


Fig. 6 Results of a single benefit assessment

Table 8 Statistical comparison results

model	MAE	RMSE	MAPE	R^2	p
traditional model	5.86	7.1	11.9%	0.83	/
Improve the model	4.21	5.18	8.3%	0.91	0.02

Table 9 The results of the β sensitivity analysis

β	Ranking consistency	Top-3 Project consistency	Top-5 Stability rate
0.1	0.96	Complete consistency	100%
0.3	1	Complete consistency	100%
0.5	0.97	Complete consistency	100%
0.7	0.95	Basically consistent	95%
0.9	0.94	Basically consistent	93%

revenue evaluation, and provides a reliable tool for guiding investment, policymaking, and the sustainable development of green power projects.

Acknowledgements This work was supported by State Grid Jibei Zhang Jiakou Wind, PV Storage and Transmission Renewable Energy CO., LTD (No. 520143230002).

Author Contributions Conceptualization, B W. and S Y; methodology, H L; software D C; validation, H L, C Y. and J K; formal analysis, C S; investigation, B W; resources, S Y; data curation, H L; writing-original draft preparation, D C; writing-review and editing, H L; visualization, C Y; supervision, J K; project ad ministration, C S; funding acquisition, B W.

Data Availability The raw data can be obtained on request from the corresponding author.

Declarations

Competing Interests The authors declare that we have no competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

1. Wang X (2017) Research on benefit evaluation of water surface photovoltaic power generation project in fishpond in B county [Master's thesis, North China Electric Power University]
2. Li X (2010) Evaluation of economic and social benefits of Baotou Beizhi 500kv project [Master's thesis, North China Electric Power University (Hebei)]
3. Du S, Liang H, Xu Y et al. (2020) Comprehensive performance evaluation of multi-index farms and stations based on composite weights. In Proceedings of the 7th China Wind Power Aftermarket Exchange and Cooperation Conference, Wind Machinery Branch of China Agricultural Machinery Industry Association & Xintian Green Energy Co. Ltd. (p. 5)
4. Xie Z, Le Q, Fan K et al. (2024, January 23). A maturity assessment method of regional green power trading market based on fuzzy comprehensive evaluation. Patent No. CN202311584505.2). Jiangxi Province
5. Yang L (2005) Research on socio-economic impact assessment of power grid construction project [Master's thesis, North China Electric Power University]
6. Han F, Ren T, Chen J (2009) Research on future evaluation of power transmission and transformation projects based on analytic hierarchy process. Electrical Technology 2:42–45
7. Yang J (2012) Research on the post-evaluation of 220KV Jindu power transmission and transformation project [Master's thesis, North China Electric Power University]

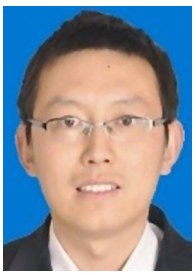
8. Zhang Y (1992) Fuzzy mathematical method and its application. China Coal Industry Press
9. Zhou K (2012) Jingyu 220KV power transmission and transformation project economic benefit evaluation research [Master's thesis, North China Electric Power University]
10. Wang Z (2013) Comprehensive evaluation study of 1000KV UHV power transmission and transformation project from Anhui to the east [Master's thesis, North China Electric Power University]
11. He H (2011) Research on the post-evaluation of economic benefits of public network power transmission and transformation project [Master's thesis, Huaqiao University]
12. Chen D, Chen M (2005) Research on economic benefit evaluation model of investment project. Shopping Mall Modernization 3:67–68
13. Hughes WR (2005) Valuing information in decision making under uncertainty. Australian Journal of Management 20(1):105–111
14. Delgado M, Herrera F (2008) Combining numerical and linguistic information in group decision making. Information Sciences 107:177–194
15. Gupta P, Harboe R, Tabucanon MT (2000) Fuzzy multi-criteria decision making for crop area planning in Narmada River basin. Agricultural Systems 63:1–18
16. Vasconcelos AGG, Rodrigues Almeida RMV et al. (2001) Path analysis and multi-criteria decision making: an approach for multivariate model selection and analysis in health. Annals of Epidemiology 11(6):377–384
17. Saaty TL (2007) Modeling unstructured decision problems-the theory of analytical hierarchies. Mathematical Modeling University of Missouri 1:90–105
18. Gregory AJ, Jackson MC (2012) Evaluation methodologies: a system of use. Journal of the Operational Research Society 1:40–51
19. Hwang CL, Yoon KL (2001) Multiple attribute decision making: methods and applications. Springer-Verlag
20. Maeda H, Murakami S (2008) A fuzzy decision-making method and its application to a company choice problem. Information Science 45(2):331–346
21. Eldukair ZA, Ayyub B (2002) Multi-attribute fuzzy decisions in construction strategies. Fuzzy Sets and Systems 42(2):155–165
22. Rogers SK (2001) Application of artificial neural networks. Wikipedia: The Free Encyclopedia 3:76–92
23. Thongkong S, Kraithong S, Singh J, Tangjaidee P, Yawootti A, Klangpetch W, Rachtanapun P, Rawdkuen S, Phongthai S (2024) Pulsed electric field for quick-cooking rice: impacts on cooking quality, physicochemical properties, and in vitro digestion kinetics. Processes 12:2577
24. Guo Y, Yu J, Yang Y, Ma H (2024, 1397) Information gap decision theory-based robust economic dispatch strategy considering the uncertainty of electric vehicles. Processes 12
25. Yan X, Zheng W, Wei Y, Yan Z (2024) Current status and economic analysis of green hydrogen energy industry chain. Processes 12:315
26. Zhao Y (2022) Research on the optimal installed capacity proportion of new energy and wind power cluster electricity allocation strategy. https://doi.org/10.26944/d.cnki.gbfju.2022.002341. [Master's thesis, Beijing Jiaotong University]
27. Shin H, Lee D, Baldick R (2017) An offer strategy for wind power producers that considers the correlation between wind power and real-time electricity prices. IEEE Transactions on Sustainable Energy 9(2):695–706
28. Zhou T, Chen Q, Zhu W, Zhang L, Kang C (2012) A comprehensive low-carbon benefits assessment model for power systems. In 2012 IEEE International Conference on Power System Technology (POWERCON), IEEE (pp. 1–5). https://doi.org/10.1109/PowerCon.2012.6401256.
29. Tang S, Zhao S, Xu Y (2021) Analysis of economic and environmental indicators for typical daily operation of regional integrated energy system. In 2021 4th International Conference on Energy, Electrical and Power Engineering (CEEPE), IEEE (pp. 1080–1090). https://doi.org/10.1109/CEEPE51765.2021.9475659
30. Dong J et al. (2024) A comprehensive assessment method for alternative energy storage in power grids based on fuzzy multi-criteria decision-making. In 2024 3rd International Conference on Energy and Electrical Power Systems (ICEEPS), IEEE (pp. 264–270). https://doi.org/10.1109/ICEEPS62542.2024.10693113
31. Mu T (2024) Research on trading strategies in the new energy retail electricity market based on artificial intelligence. Electronic Components and Information Technology 8(12):80–83. https://doi.org/10.19772/j.cnki.2096-4455.2024.12.026
32. Han J, Wang J, He Z et al. (2023) Hydrogen-powered smart grid resilience. Energy Conversion and Economics 4(2):89–104
33. Ding Z, Wang J, Song Y, Zheng X, He G, Chen X, Zhang T, Lee W-J, Song J (2025) Tracking the carbon footprint of global generative Artificial Intelligence. The Innovation 6:100866. https://doi.org/10.1016/j.xinn.2025.100866
34. Song Y, Wang J, Zhang T, He G, Wang Y (2025) Characterizing spatiotemporal-coupled resilience of power systems via a security region-based method. CSEE Journal of Power and Energy Systems, early access
35. Ding Z, Wang J, Song Y, Zheng X, He G, Chen X, Zhang T, Lee W-J, Song J (2025) Tracking the carbon footprint of global generative Artificial Intelligence. The Innovation 6(5)

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.

Bin Wang He graduated from Northeast Electric Power University and is a Senior Engineer. His research focuses on electricity markets, power system optimization, and new energy regulation and operation.

Shuailiang Yao He graduated from Hebei University of Technology and is an Engineer. His research focuses on electricity markets, power system optimization, and new energy regulation and operation.



Haixu Liu He graduated from Huaqiao University and is a Senior Engineer. His research focuses on financial management.

Dongxia Cheng He graduated from North China University of Water Resources and Electric Power and is a Senior Engineer.His research focuses on human resources.

Hongyong Liu He graduated from Hebei Agricultural University and is an Engineer. His research focuses on electricity markets, power system optimization, and new energy regulation and operation.

Jian Kou He graduated from Hebei Agricultural University with a bachelor's degree and is a Senior Engineer.His research focuses on communication automation.

Chengdong Yang He graduated from Lanzhou University of Technology and is an Engineer. His research focuses on substation operation and maintenance.

Changjiang Su He graduated from Hebei Agricultural University and is an Engineer. His research focuses on information and communication.